Top 24 CAT Maxima-Minima Questions With Video Solutions

Questions

Instructions

Ganga, Kaveri, and Narmada are three women who buy four raw materials (Mango, Apple, Banana and Milk) and sell five finished products (Mango smoothie, Apple smoothie, Banana smoothie, Mixed fruit smoothie and Fruit salad). Table-1 gives information about the raw materials required to produce the five finished products. One unit of a finished product requires one unit of each of the raw materials mentioned in the second column of the table.

Table-1

Finished product	Raw materials required
Mango smoothie	Mango, Milk
Apple smoothie	Apple, Milk
Banana smoothie	Banana, Milk
Mixed fruit smoothie	Mango, Apple, Banana, Milk
Fruit salad	Mango, Apple, Banana

One unit of milk, mango, apple, and banana cost ₹5, ₹3, ₹2, and ₹1 respectively. Each unit of a finished product is sold for a profit equal to two times the number of raw materials used to make that product. For example, apple smoothie is made with two raw materials (apple and milk) and will be sold for a profit of ₹4 per unit. Leftover raw materials are sold during the last business hour of the day for a loss of ₹1 per unit.

The amount, in rupees, received from sales (revenue) for each woman in each of the four business hours of the day is given in Table-2.

Table-2

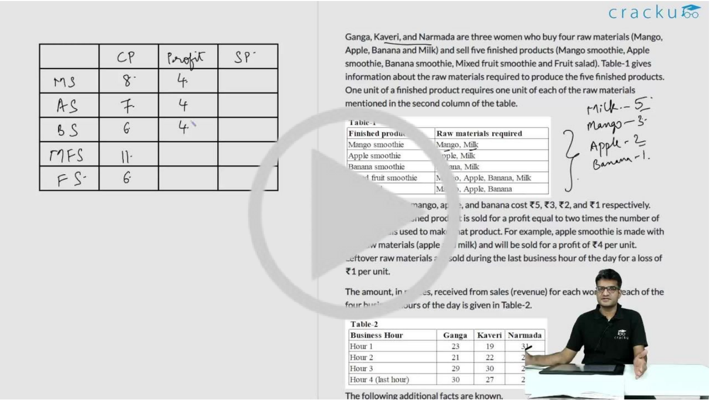
Business Hour	Ganga	Kaveri	Narmada
Hour 1	23	19	31
Hour 2	21	22	21
Hour 3	29	30	23
Hour 4 (last hour)	30	27	22

The following additional facts are known.
1. No one except possibly Ganga sold any Mango smoothie.
2. Each woman sold either zero or one unit of any single finished product in any hour.
3. Each woman had exactly one unit each of two different raw materials as leftovers.
4. No one had any banana leftover.

Question 1

What BEST can be concluded about the number of units of fruit salad sold in the first hour?

- **A** Either 1 or 2.
- **B** Either 0 or 1 or 2.
- **C** Exactly 2.
- **D** Exactly 1.



VIDEO SOLUTION

Question 2

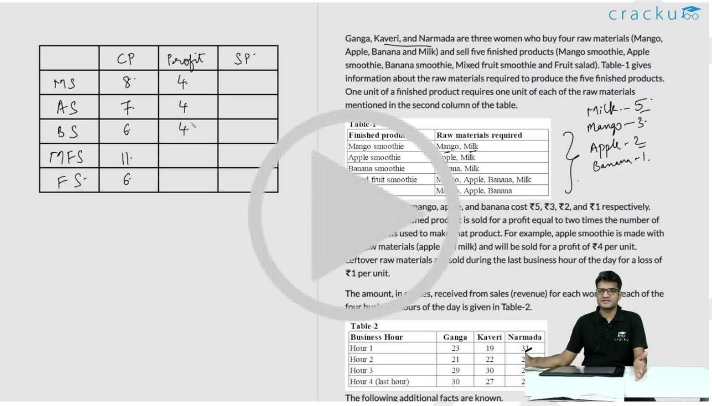
Which of the following is NECESSARILY true?

A Ganga did not sell any leftover mangoes.

B Ganga did not sell any leftover apples.

C Narmada sold one unit of leftover milk.

D Kaveri sold one unit of leftover mangoes.



VIDEO SOLUTION

Question 3

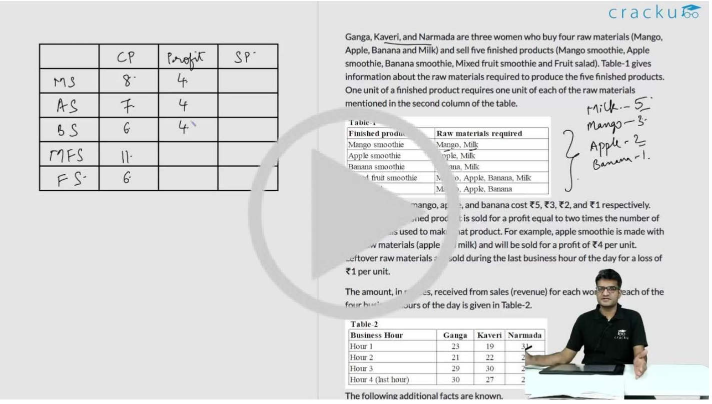
What BEST can be concluded about the total number of units of milk the three women had in the beginning?

A Either 19 or 20 units.

B Either 18 or 19 or 20 units.

C Either 18 or 19 units.

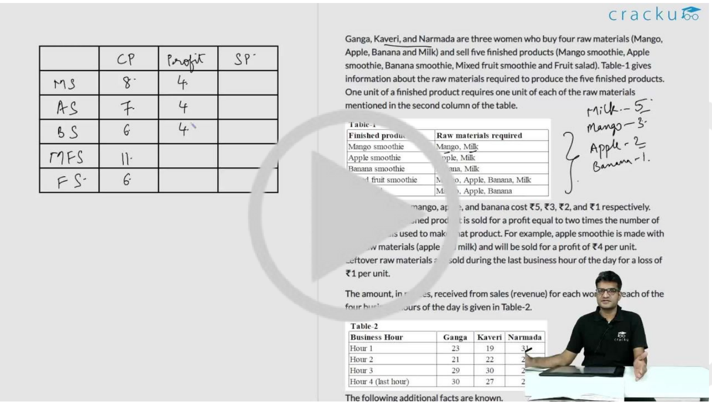
D Either 17 or 18 or 19 units.



VIDEO SOLUTION

Question 4

If it is known that three leftover units of mangoes were sold during the last business hour of the day, how many apple smoothies were sold during the day?



VIDEO SOLUTION

Instructions

Ravi works in an online food-delivery company. After each delivery, customers rate Ravi on each of four parameters - Behaviour, Packaging, Hygiene, and Timeliness, on a scale from 1 to 9. If the total of the four rating points is 25 or more, then Ravi gets a bonus of ₹20 for that delivery. Additionally, a customer may or may not give Ravi a tip. If the customer gives a tip, it is either ₹30 or ₹50.

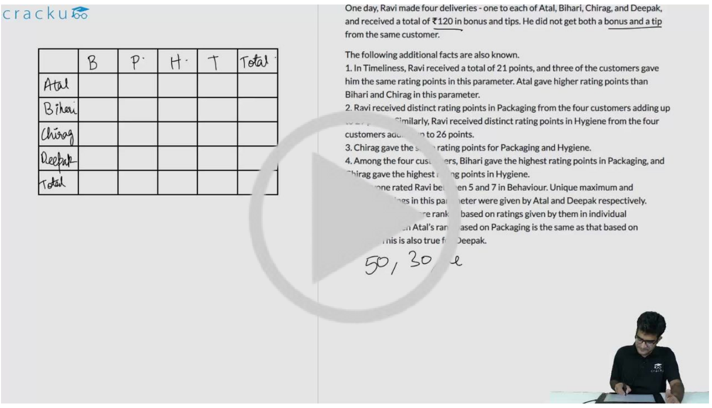
One day, Ravi made four deliveries - one to each of Atal, Bihari, Chirag, and Deepak, and received a total of ₹120 in bonus and tips. He did not get both a bonus and a tip from the same customer.

The following additional facts are also known.
1. In Timeliness, Ravi received a total of 21 points, and three of the customers gave him the same rating points in this parameter. Atal gave higher rating points than Bihari and Chirag in this parameter.
2. Ravi received distinct rating points in Packaging from the four customers adding up to 29 points. Similarly, Ravi received distinct rating points in Hygiene from the four customers adding up to 26 points.
3. Chirag gave the same rating points for Packaging and Hygiene.
4. Among the four customers, Bihari gave the highest rating points in Packaging, and Chirag gave the highest rating points in Hygiene.

5. Everyone rated Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.
6. If the customers are ranked based on ratings given by them in individual parameters, then Atal's rank based on Packaging is the same as that based on Hygiene. This is also true for Deepak.

Question 5

What was the minimum rating that Ravi received from any customer in any parameter?



VIDEO SOLUTION

Question 6

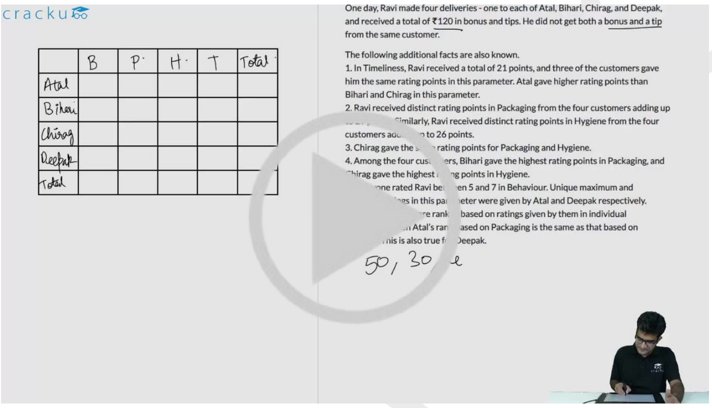
What rating did Deepak give on Packaging?

A 7

B 8

C 5

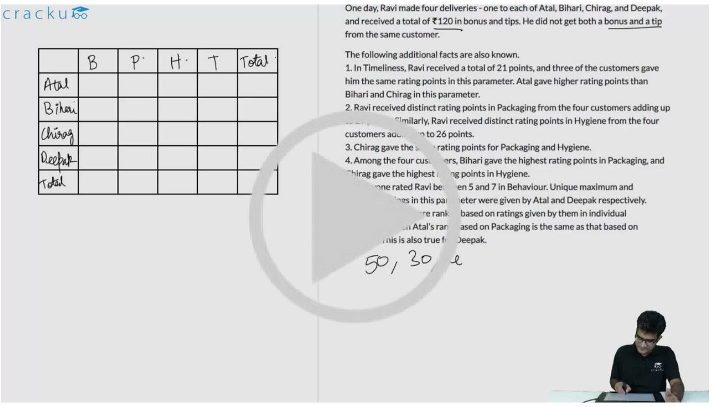
D 6



VIDEO SOLUTION

Question 7

What rating did Atal give on Timeliness?



VIDEO SOLUTION

Question 8

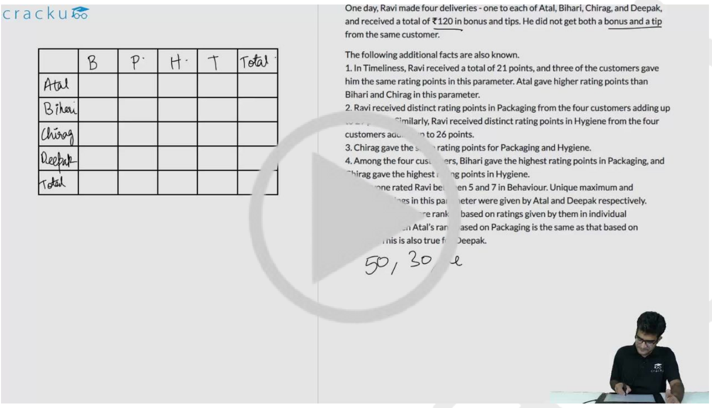
In which parameter did Atal give the maximum rating points to Ravi?

A Hygiene

B Behaviour

C Timeliness

D Packaging



VIDEO SOLUTION

Question 9

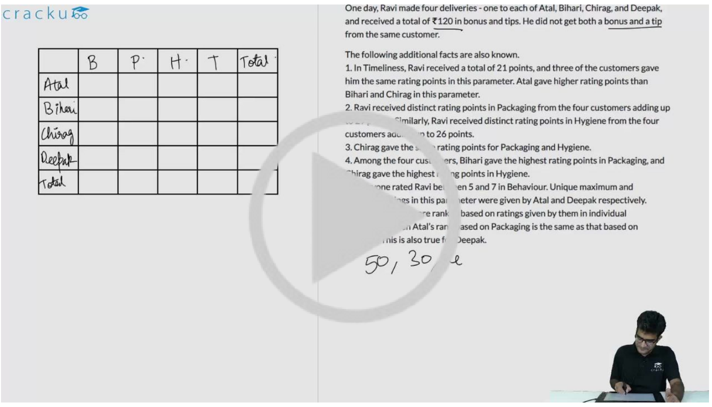
What BEST can be concluded about the tip amount given by Deepak?

A Either ₹0 or ₹30 or ₹50

B Either ₹30 or ₹50

C ₹50

D ₹30



VIDEO SOLUTION

Question 10

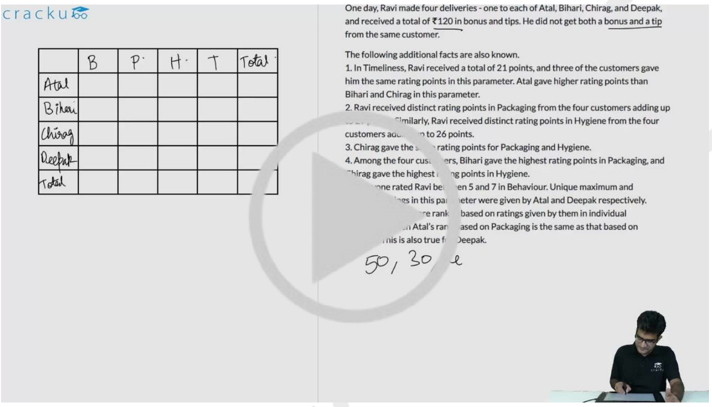
The COMPLETE list of customers who gave the maximum total rating points to Ravi is

A Atal

B Bihari

C Bihari and Chirag

D Atal and Bihari



VIDEO SOLUTION

Instructions

The local office of the APP-CAB company evaluates the performance of five cab drivers, Arun, Barun, Chandan, Damodaran, and Eman for their monthly payment based on ratings in five different parameters (P1 to P5) as given below:
P1: timely arrival
P2: behaviour
P3: comfortable ride
P4: driver's familiarity with the route
P5: value for money

Based on feedback from the customers, the office assigns a rating from 1 to 5 in each of these parameters. Each rating is an integer from a low value of 1 to a high value of 5. The final rating of a driver is the average of his ratings in these five parameters. The monthly payment of the drivers has two parts - a fixed payment and final rating-based bonus. If a driver gets a rating of 1 in any of the parameters, he is not eligible to get bonus. To be eligible for bonus a driver also needs to get a rating of five in at least one of the parameters. The partial information related to the ratings of the drivers in different parameters and the monthly payment structure (in rupees) is given in the table below:

	P1	P2	P3	P4	P5	Fixed Payment	Bonus
Arun				4		Rs.1000	Rs.250× Final Rating
Barun	3					Rs.1200	Rs.200× Final Rating
Chandan			2			Rs.1400	Rs.100× Final Rating
Damodaran		3				Rs.1300	Rs.150× Final Rating
Eman					2	Rs.1100	Rs.200× Final Rating

The following additional facts are known.
1. Arun and Barun have got a rating of 5 in exactly one of the parameters. Chandan has got a rating of 5 in exactly two parameters.
2. None of drivers has got the same rating in three parameters.

Question 11

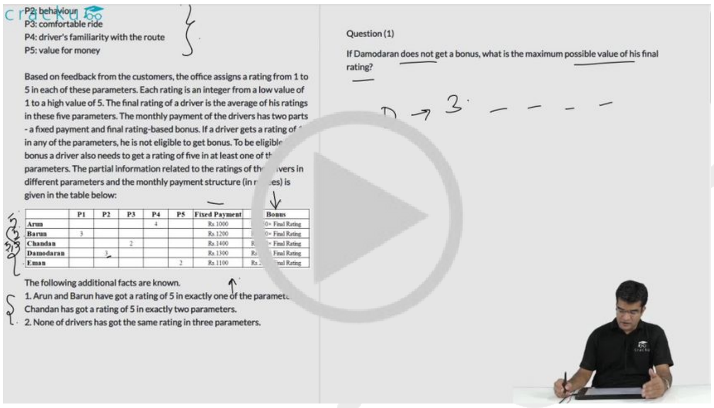
If Damodaran does not get a bonus, what is the maximum possible value of his final rating?

A 3.4

B 3.2

C 3.6

D 3.8



VIDEO SOLUTION

Question 12

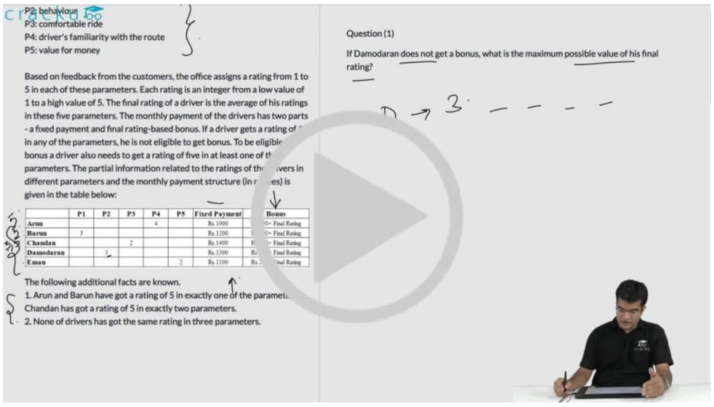
If Eman gets a bonus, what is the minimum possible value of his final rating?

A 3.2

B 2.8

C 3.4

D 3.0



VIDEO SOLUTION

Question 13

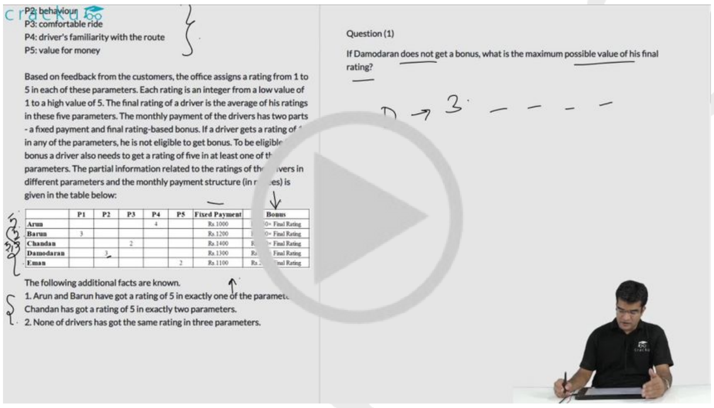
If all five drivers get bonus, what is the minimum possible value of the monthly payment (in rupees) that a driver gets?

A 1750

B 1600

C 1740

D 1700



VIDEO SOLUTION

Question 14

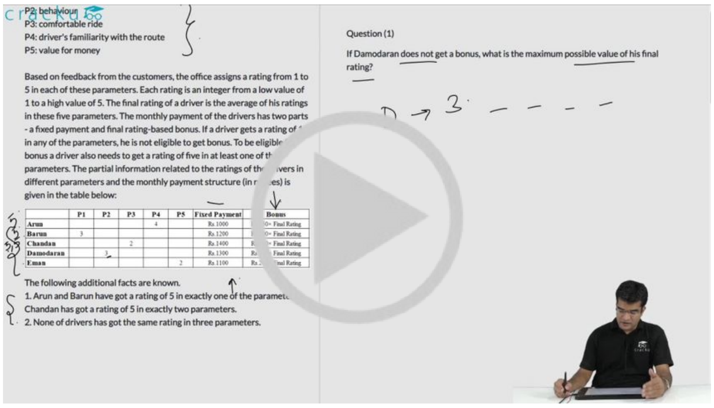
If all five drivers get bonus, what is the maximum possible value of the monthly payment (in rupees) that a driver gets?

A 1960

B 2050

C 1950

D 1900



VIDEO SOLUTION

Instructions

Each of the bottles mentioned in this question contains 50 ml of liquid. The liquid in any bottle can be 100% pure content (P) or can have certain amount of impurity (I). Visually it is not possible to distinguish between P and I. There is a testing device which detects impurity, as long as the percentage of impurity in the content tested is 10% or more.

For example, suppose bottle 1 contains only P, and bottle 2 contains 80% P and 20% I. If content from bottle 1 is tested, it will be found out that it contains only P. If content of bottle 2 is tested, the test will reveal that it contains some amount of I. If 10 ml of content from bottle 1 is mixed with 20 ml content from bottle 2, the test will show that the mixture has impurity, and hence we can conclude that at least one of the two bottles has I. However, if 10 ml of content from bottle 1 is mixed with 5 ml of content from bottle 2. the test will not detect any impurity in the resultant mixture.

Question 15

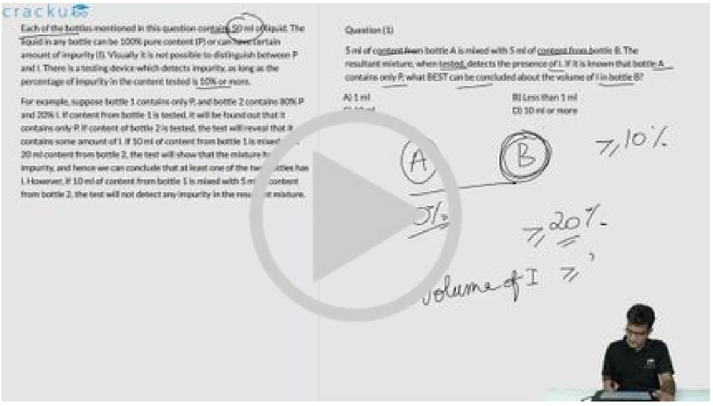
5 ml of content from bottle A is mixed with 5 ml of content from bottle B. The resultant mixture, when tested, detects the presence of I. If it is known that bottle A contains only P, what BEST can be concluded about the volume of I in bottle B?

A 1 ml

B Less than 1 ml

C 10 ml

D 10 ml or more



VIDEO SOLUTION

Question 16

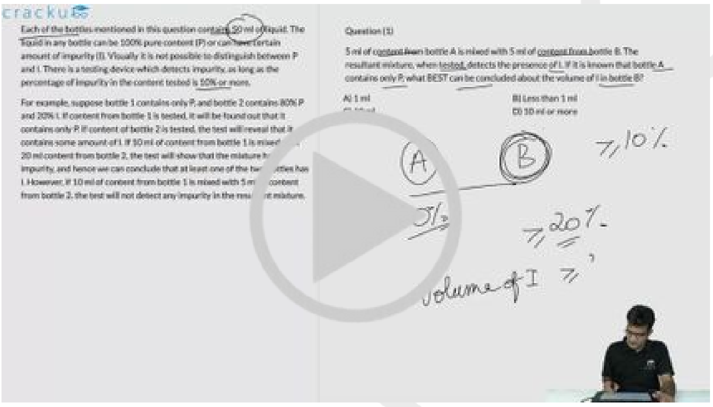
There are four bottles. It is known that either one or two of these bottles contain(s) only P, while the remaining ones contain 85% P and 15% I. What is the minimum number of tests required to ascertain the exact number of bottles containing only P?

A 4

B 2

C 3

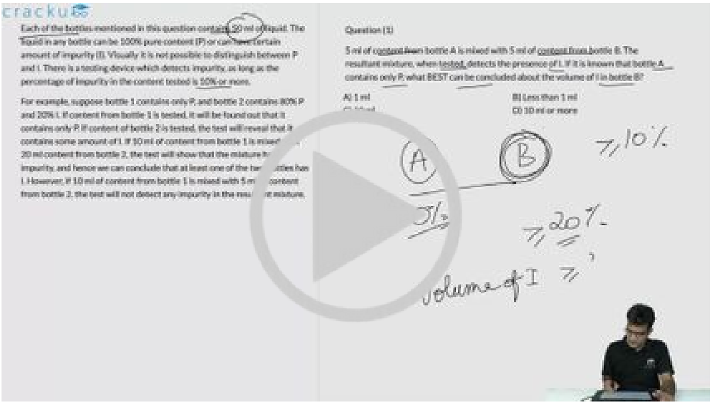
D 1



VIDEO SOLUTION

Question 17

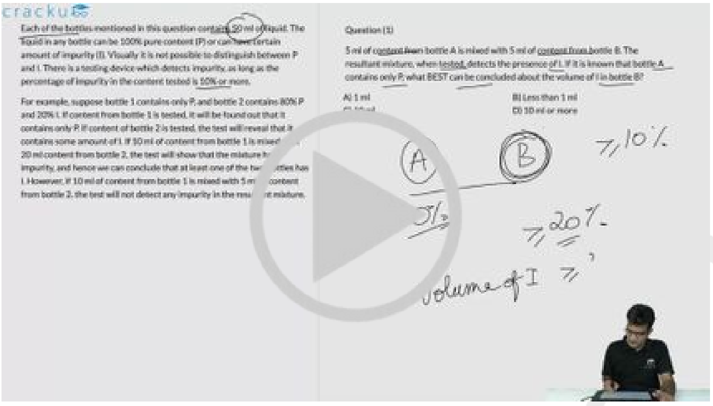
There are four bottles. It is known that three of these bottles contain only P, while the remaining one contains 80% P and 20% I. What is the minimum number of tests required to definitely identify the bottle containing some amount of I?



VIDEO SOLUTION

Question 18

There are four bottles. Each bottle is known to contain only P or only I. They will be considered to be "collectively ready for despatch" if all of them contain only P. In minimum how many tests, is it possible to ascertain whether these four bottles are "collectively ready for despatch"?



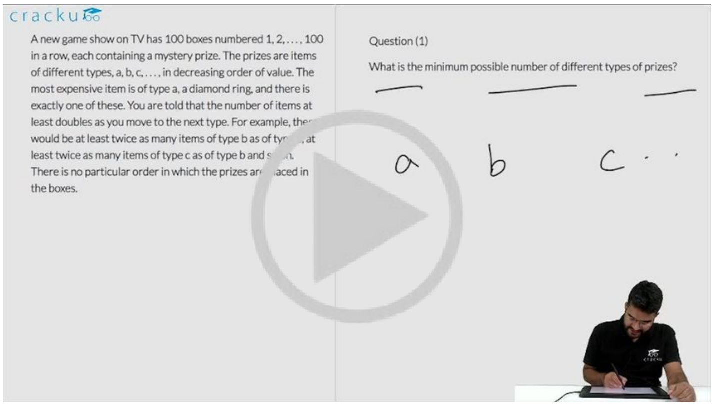
VIDEO SOLUTION

Instructions

A new game show on TV has 100 boxes numbered 1, 2, . . . , 100 in a row, each containing a mystery prize. The prizes are items of different types, a, b, c, . . . , in decreasing order of value. The most expensive item is of type a, a diamond ring, and there is exactly one of these. You are told that the number of items at least doubles as you move to the next type. For example, there would be at least twice as many items of type b as of type a, at least twice as many items of type c as of type b and so on. There is no particular order in which the prizes are placed in the boxes.

Question 19

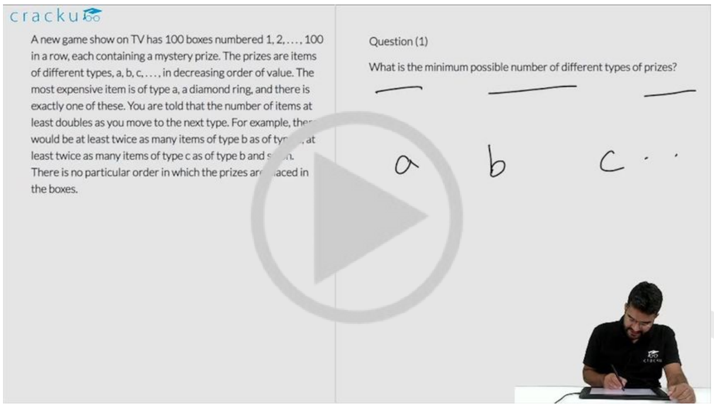
What is the minimum possible number of different types of prizes?



VIDEO SOLUTION

Question 20

What is the maximum possible number of different types of prizes?



VIDEO SOLUTION

Question 21

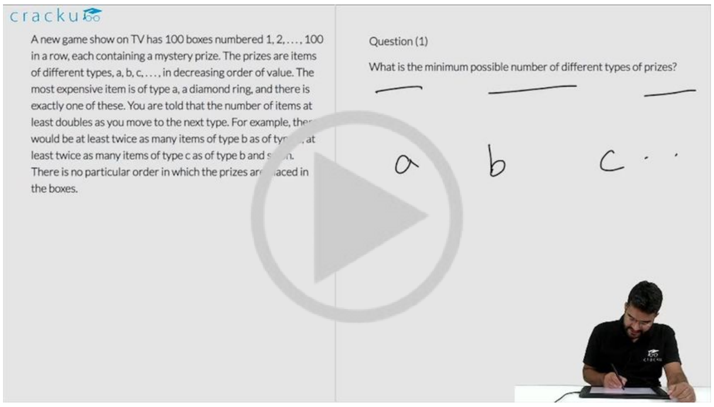
Which of the following is not possible?

A There are exactly 75 items of type e.

B There are exactly 30 items of type b.

C There are exactly 45 items of type c.

D There are exactly 60 items of type d.



VIDEO SOLUTION

Question 22

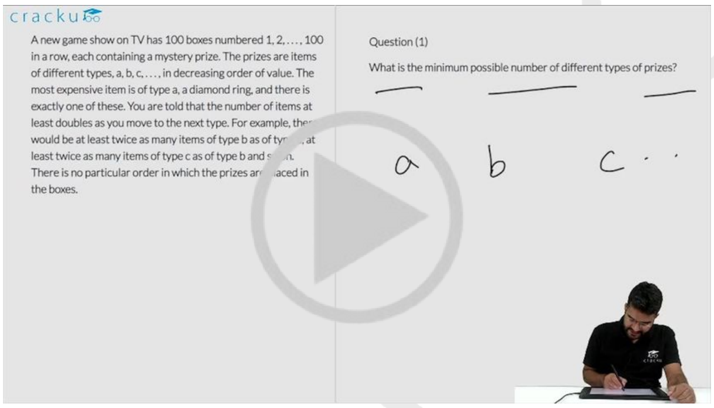
You ask for the type of item in box 45. Instead of being given a direct answer, you are told that there are 31 items of the same type as box 45 in boxes 1 to 44 and 43 items of the same type as box 45 in boxes 46 to 100. What is the maximum possible number of different types of items?

A 5

B 6

C 4

D 3



VIDEO SOLUTION

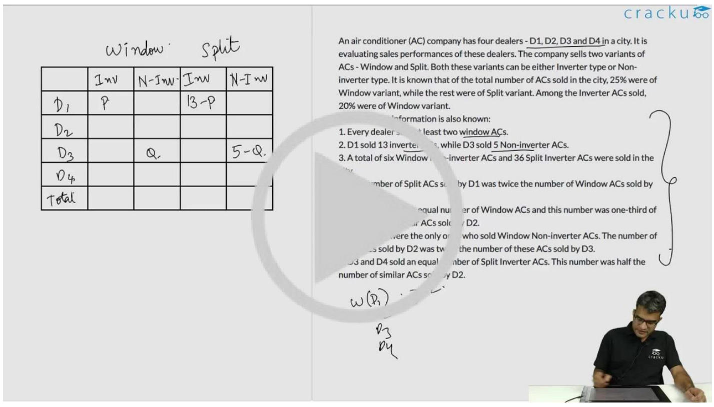
Instructions

An air conditioner (AC) company has four dealers - D1, D2, D3 and D4 in a city. It is evaluating sales performances of these dealers. The company sells two variants of ACs - Window and Split. Both these variants can be either Inverter type or Non-inverter type. It is known that of the total number of ACs sold in the city, 25% were of Window variant, while the rest were of Split variant. Among the Inverter ACs sold, 20% were of Window variant.
The following information is also known:

1. Every dealer sold at least two window ACs.
2. D1 sold 13 inverter ACs, while D3 sold 5 Non-inverter ACs.
3. A total of six Window Non-inverter ACs and 36 Split Inverter ACs were sold in the city.
4. The number of Split ACs sold by D1 was twice the number of Window ACs sold by it.
5. D3 and D4 sold an equal number of Window ACs and this number was one-third of the number of similar ACs sold by D2.
6. D2 and D3 were the only ones who sold Window Non-inverter ACs. The number of these ACs sold by D2 was twice the number of these ACs sold by D3.
7. D3 and D4 sold an equal number of Split Inverter ACs. This number was half the number of similar ACs sold by D2.

Question 23

What was the total number of ACs sold by D2 and D4?



VIDEO SOLUTION

Question 24

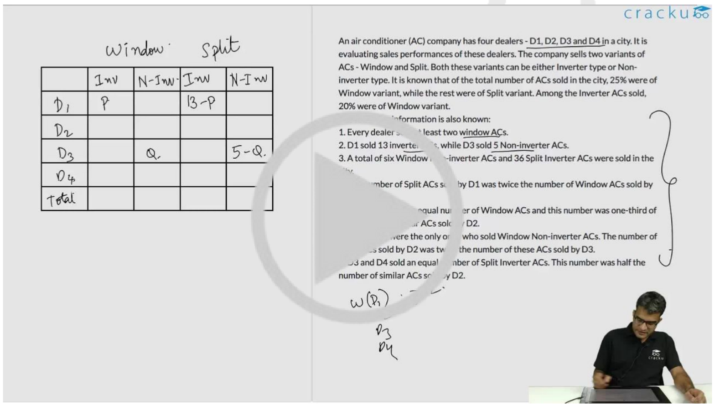
What percentage of ACs sold were of Non-inverter type?

A 33.33%

B 75.00%

C 25.00%

D 20.00%



VIDEO SOLUTION